

Three good links, one edge each

Even Step, In Step and Short Run, against each other and against today's link · every figure computed from the rotas

What all three share

Fifteen designs are in the pack; these three balance best what weighs most here. Everything below holds for all three, each line better than or level with today; what they share that is worse than today is on page 2.

✓ All **9 of 9** December staffing rules (today 4), and all three flexible ones: 14 on a Saturday, evenly spaced cover weeks, weekday closers from 15:45

✓ **7 full weekends off in 26**, never more than 5 weeks apart (today 4 in 20, up to 7 apart)

✓ Every closing shift **shorter, by 30 minutes to an hour** — the weekday's and Sunday's 8h 10m on all three (today 8h 40m and 8h 55m); no shift over 9h 00m

✓ **4 single rest days** in the rotation, as today — about 8 a year each, today 10; every other break is two days (22 of 26; today 13 of 17 are two or more, five of them three or four)

✓ **No avoidable fatigue warning** on the fixed rota, one if a cover week falls badly — a 55-hour week; never more than **five days in a row** on the fixed rota, seven if a cover week falls badly (today 4 warnings, up to 5; seven days, up to nine)

✓ The same **35-hour** week and **218.6 days** a year, under the ceiling of 219 (today 219.0)

✓ **One shift time Monday to Friday and no early–late switch in 18 of 21 weeks**, no week mixing the two (today 7 of 16, with 7 mixing)

✓ **The same Sunday**: the same ten duties, one of them a 9-hour early, 09:00–18:00

Where they differ

the better of the three on each line in bold green; today for scale, not ranked

MEASURE	TODAY	EVEN STEP	IN STEP	SHORT RUN
Heaviest week, Mon–Sat	43h 50m	41h 30m	**41h 05m**	42h 30m
Lightest week, Mon–Sat	25h 10m	**25h 30m**	**25h 30m**	24h 30m
Shortest rest, fixed duties	12h 30m	**14h 35m**	**14h 35m**	14h 20m
Closing shift, Saturday	9h 10m	**8h 10m**	**8h 10m**	8h 40m
Earlies over 8h 30m, Mon–Sat, each a year	0	**0**	40	20
Any shift over 8h 30m, Mon–Sat, each a year	31	**0**	40	26
Earlies finishing before 14:00, each a year	26	10	**12**	0
Weekday lates, closers aside, on average	8h 30m	8h 27m	8h 12m	**8h 00m**
Week-to-week change of start	4h 00m	1h 31m	**1h 13m**	**1h 13m**
Shift times, and how many are new	18 (0 new)	**13 (5 new)**	15 (6 new)	15 (6 new)
New shift times on a Saturday	0	1	1	**0**
Fit to the trains, weekday · Sat (0 = exact; Sunday 29.0 on all three, today 71.7)	52.0 · 65.9	32.6 · 10.0	29.8 · **9.6**	**29.1** · 12.7

Each one's edge

Even Step ES-26-T5 · d1921b7a

No long days. No shift over 8h 30m and every early one of today's times, Monday to Saturday; In Step's 8h 10m closers, today's 13:45, and the fewest shift times of the three, 13. Its cost is the weekday lates, 8h 27m on average, near today's 8h 30m (In Step's 8h 12m), the weakest weekday fit (32.6) and a 1h 31m change of start.

In Step IS-26-T4 · 8d9239f3

Steady hours, early finishes. Weeks of 25h 30m to 41h 05m, Mon–Sat, 12 finishes a year each before 14:00, the best Saturday fit of the three (9.6), and Even Step's closers and 13:45. Its cost is the earlies: 40 a year each over 8h 30m, Monday to Saturday, at 8h 45m and 8h 55m — the most of any design here.

Short Run SR-26-F1 · 618348d6

The closest weekday fit, the shortest weekday lates. 29.1 against 29.8 and 32.6, weekday lates of 8h 00m on average, and nothing new on a Saturday. Its cost: an 8h 40m Saturday closer, a 42h 30m heaviest week, 14h 20m the shortest rest, no early finish before 14:00, and 20 earlies a year over 8h 30m, all 9h 00m.

Which one, by what matters most

IF THE MAIN PRIORITY IS	CHOOSE
Of these three, no shift over 8h 30m from Monday to Saturday	**Even Step**
Of these three, the steadiest weekly hours and the most early finishes	**In Step**
Of these three, the closest weekday fit and the shortest weekday lates	**Short Run**
The shortest closers of all: 16:00 Monday to Saturday	**Closing Time** · outside the shortlist, as is Both Ways
The fewest new shift times: two	**Home Grown** · outside the shortlist
A fourth person to the Saturday close	**Even Keel** · outside the shortlist, as are Fine Tune and Second Gear
The steadiest start times from week to week	**Second Edition** · outside the shortlist
Four weeks off for 14 days' leave, as today	**Long Break** · outside the shortlist

THE SHORTLIST · CONTINUED

Against today, and a recommendation

On most of what a colleague feels the three are the same — a figure in grey is shared by all three

Against today's link

MEASURE	TODAY	EVEN STEP	IN STEP	SHORT RUN
Full weekends off, about a year	10	14	14	14
Saturdays a year, each	26	28	28	28
Sunday overtime duties a year, each	21	20	20	20
Starting at 06:20, each a year	62	60	60	60
Average shift, Monday to Saturday	8h 14m	8h 16m	8h 16m	8h 16m
Longest shift	9h 10m	9h 00m	9h 00m	9h 00m
14 days' leave buys, at best	30 days off	28 days off	28 days off	28 days off
14 days' leave buys, at worst	19 days off	20 days off	20 days off	20 days off
Leave for four full weeks off	14 days	15 days	15 days	15 days

Per year: the rotation's count scaled to 52 weeks; days a year, Monday to Saturday with a cover week as four, over 365 days.

Why these three and not the other twelve

These three balance best what weighs most here — days in a row, weekends, one shift time a week, late finishes and the closing shift. Each of the others keeps something none of the three has, and pays for it. Closing Time and Both Ways have shorter closers — Closing Time also more even weeks, a gentler step and the best Saturday fit; Both Ways a lighter heaviest week, more early finishes and fewer new times — for fewer weeks on one turn, and a weaker weekday fit and more finishes at 22:00 or later on Closing Time, more long shifts and a weaker weekend fit on Both Ways. Home Grown has the fewest new times, two, no early over 8h 30m on any day and a shorter weekday closer, for today's 9h 10m Saturday closer, fewer weeks on one turn and a weaker fit. Even Keel, Fine Tune and Second Gear have a fourth person to the Saturday close (Fine Tune a gentler step), for weeks mixing earlies and lates, longer blocks, more late finishes and a weaker Saturday fit, and on Fine Tune and Second Gear fewer weeks on one turn. Second Edition has the gentlest step, for six days in a row. Long Break gives four weeks off for 14 days' leave and a 29-day best stretch, for six days in a row, a fifth single rest day, fewer weeks on one turn and a 1h 44m step. Second Look, Second Nature, Second Sight and Second Wind have blocks a week shorter (Second Look also six days in a row at worst, not seven), for six days in a row on the fixed rota and fewer weeks on one turn — Second Sight and Second Wind a bigger step, Second Nature weekends up to ten weeks apart.

The honest costs, shared by all three

- **Late finishes.** 42 a year each at 23:00 or later (today 39) and 94 at 22:00 or later (today 78) on all three. Three to the close every day, where today has two Monday to Saturday, means no link meeting the rules has fewer than 42 at 23:00 or later.
- **Early finishes.** Fewer before 14:00 than today's 26 a year each: 10 on Even Step, 12 on In Step, none on Short Run.
- **Leave.** The best 14-day stretch is 28 days off, not today's 30; four full weeks off, Sunday to Saturday, takes 15 days' leave, not 14. The worst stretch is a day better. All three are worse than today here, on something people plan a year around; Long Break alone keeps the 14.
- **Saturdays.** About 2 more a year (28, today 26): fourteen on a Saturday, the flexible rule, across 26 lines (today ten across 20).
- **New times.** 5 of Even Step's 13 shift times are new, 6 of 15 on the other two: on Even Step three shorter versions of today's lates, a 12:15–20:30 and the Sunday early; on In Step three shorter lates and two longer weekday earlies; on Short Run four and one. All three add the 9-hour Sunday early, 09:00–18:00.

Recommendation

> **Even Step.** In Step's weeks without the long earlies. On what weighs most here it equals both — days in a row, weekends, weeks on one shift time, late finishes and the leave — with In Step's closers, every one Monday to Saturday 8h 10m (Short Run's Saturday one 8h 40m). Monday to Saturday it alone has no shift over 8h 30m (In Step 40 a year each, Short Run 26, today 31) and every early is one of today's times; the 13:45 stays every weekday, and it has the fewest shift times of the three, 13 with 5 new. It equals In Step's lightest week and 14h 35m rest; its heaviest week (41h 30m), early finishes (10 a year) and Saturday fit (10.0) sit between the other two. Its costs: the longest weekday lates, 8h 27m on average (In Step's 8h 12m, Short Run's 8h 00m), the weakest weekday fit (32.6, against 29.8 and 29.1), a 1h 31m change of start (the others 1h 13m) and, as on In Step, a new Saturday time, its closer.

But it is a judgement. In the pack's weighted league (the owner's weights, which count long earlies and the closing shift since 31 October) Even Step is first of the fifteen, on 84.0, just ahead of Closing Time (83.8); Short Run is seventh and In Step eighth. Weighted as it was before, In Step led, under a point ahead of Even Step (87.4 against 86.6). So choose by what matters here:

- **Choose Even Step** if long days matter most — for lates near today's 8h 30m, the weakest weekday fit and a 1h 31m change of start.
- **Choose In Step** if, on the same weeks, short lates matter more than short earlies — for 40 earlies a year each over 8h 30m.
- **Choose Short Run** if the weekday fit and short weekday lates matter most — for an 8h 40m Saturday closer, a 42h 30m Mon–Sat week, no 13:45 and 20 long earlies a year.